

Replication and reproduction

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Methods results and limitations

Without opening the old lesson, explain one key idea from Methods results and limitations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Reading a research claim

Without opening the old lesson, explain one key idea from Reading a research claim. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Replication and reproduction. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Replication and reproduction. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Team repeats old six queries, then evaluates four newly reserved queries. How many distinct cases and which are new evidence?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Partner reruns the same eight cases and gets the same score. Team counts sixteen independent cases. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Exact replay can preserve the original design flaws; new data can change conditions and needs its own record. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Team repeats old six queries, then evaluates four newly reserved queries. How many distinct cases and which are new evidence?

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled replication and reproduction evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of replication and reproduction. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Replication and reproduction

Same records versus new experiment. A team first reruns a supplied analysis, then collects new records with the same protocol.

Original recorded labels: [1,0,1,1]. Analysis counts correct cases. New independent set: [1,1,0,0]. Binary entries mean 1=correct and 0=incorrect.

AI ACADEMY / AI-7/5

Different four-record datasets

correct cases out of 4; samples differ

Saved analysis



New sample



Bar length shows the supplied value; the marked baseline is zero.

Replication and reproduction

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Reproduce the analysis

Reusing the supplied four records gives $3/4=75\%$.



2. Repeat the protocol

The new four-record set gives $2/4=50\%$.



3. Compare purposes

The first checks computational reproduction; the second probes repeatability with new observations.

CHANGE ONE THING

The rerun of the original records outputs $2/4$.

Explain the changed result using the matching lesson.

Calculate the scores from the original and new four-record sets. Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Same records versus new experiment. A team first reruns a supplied analysis, then collects new records with the same protocol.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The first checks computational reproduction; the second probes repeatability with new observations.

1. Compare purposes
2. Reproduce the analysis
3. Repeat the protocol

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compare purposes The first checks computational reproduction; the second probes repeatability with new observations. Reusing data and collecting new data answer different questions. A different new-sample result can arise without a code error; inspect both protocols and variability.

Practice 2: explain the deciding evidence

Changed-case practice: The rerun of the original records outputs 2/4. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. Investigate code, version, input or scoring differences.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Investigate code, version, input or scoring differences. The same fixed records should give the same deterministic count.

Repair a missing building block

Review Methods results and limitations (ai-7/4). Goal: Explain the mechanism and terminology of methods results and limitations. Short practice: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor evidence in their proper categories. Why cannot the result establish that B is best for outdoor farming?. Return to this week and explain what you changed.

Review Reading a research claim (ai-7/2). Goal: Explain the mechanism and terminology of reading a research claim. Short practice: Calculate the baseline and new rates for 2/10 and 4/10. Explain both twice the rate and the percentage-point change, and say why neither measures every learner's improvement.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
